Inyoung Oh

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Google Scholar · GitHub

Education

Ph.D. in Mechanical and Robotics Engineering / M.S. in Mechatronics / GIST — **Feb 2026** (Ph.D.) / **Feb 2018** (M.S.)

- Ph.D. Dissertation - "Normal Vector Estimation and Semantic Segmentation of 3D Point Clouds using Deep Learning and Geometric Analysis" (Grade: 4.2/4.5)
- M.S. Thesis - "Sphere and Cylinder Detection in Kinect Point Clouds using RANSAC and the 2D Hough Transform" (Grade: 3.91/4.5)

B.S. in Mechatronics / CNU (Grade: 3.913/4.5) — **Feb 2016** (B.S.)
Chungnam National University, Daejeon, Republic of Korea

Research Experience (Advisor: Prof. Kwang Hee Ko)

Geometry-Aware 3D Perception and Representation Learning

- Designed SFD-Net for sharp feature detection using multi-scale geometric descriptors with transformer attention, achieving SOTA F1-score on ABC and transferring to Kinect v2/S3DIS.
- Developed a multi-scale attention-based normal estimation with noise-aware supervision and geometric regularization, achieving SOTA on the PCPNet primitive subset.
- Developed a parameter-free curvature-driven cylinder recognition method.

LiDAR Perception for Autonomous Driving

- Developed an intensity-assisted normal-aware LiDAR semantic segmentation module, improving SemanticKITTI mIoU by +4% and boosting safety-critical classes (Bicycles +21%, Pedestrians +4%, Motorcyclists +15.2%).

2D Image-Based 6-DoF Pose Estimation and Mixed Reality Systems

- Developed RoI-PCA color-space augmentation for robust 6-DoF pose estimation from 2D images and deployed it in a real-time MR remote collaboration system.

Publications

A Mixed Reality-based Remote Collaboration Framework Using Improved Pose Estimation
Inyoung Oh; Gilsang Jang; Jinho Song; Moongu Son; Daewoon Kim; Junsang Yun; Kwang Hee Ko.
Computers in Industry, vol. 174, 104414, **2026**.

SFD-Net: Sharp Feature Detection Network Based on Local Geometric Features
Inyoung Oh; Kwang Hee Ko.
Under review at *Journal of Computational Design and Engineering*; submitted in **Nov 2025**.

Enhanced Point Cloud Normal Estimation via Multi-Scale Geometric Feature Fusion and Attention Mechanism
Inyoung Oh; Minsung Kim; Dongho Yun; Kwang Hee Ko.
Under review at *International Journal of Image and Graphics*; submitted in **Nov 2025**.

Improved semantic segmentation network using normal vector guidance for LiDAR point clouds
Minsung Kim; Inyoung Oh; Dongho Yun; Kwang Hee Ko.
Journal of Computational Design and Engineering, vol. 10, issue 6, 2332–2344, **2023**.

Automated recognition of 3D pipelines from point clouds
Inyoung Oh; Kwang Hee Ko.
The Visual Computer, vol. 37, issue 6, 1385–1400, **2021**.

Presentations

Automatic detection of cylindrical objects from unorganized point cloud **2018**
Inyoung Oh; Kwang Hee Ko.
EuroVR (poster), London, UK.

A RANSAC-based Method for Detection of Multiple Spheres from a Point Cloud **2017**
Inyoung Oh; Kwang Hee Ko.
Computer Graphics International (poster), Yokohama, Japan.

Patents

Automated System for Detecting 3D Pipeline
Kwang Hee Ko; Inyoung Oh.
KR 2010-2135828 B1 (Registered), **July 8, 2020**.

Mentorship and Teaching Experiences

Mentorship, Modeling and Simulation Lab **2018–Present**
Mentored interns through end-to-end experimental workflows and data collection, leading to successful completion of a bachelor's thesis.

Teaching Assistant, Engineering Analysis **2020**
Led weekly recitations/office hours and designed assessment materials while delivering biweekly Python instruction to strengthen quantitative problem-solving.

Teaching Assistant, Engineering Analysis **2017**
Delivered weekly instruction and assessment support, including biweekly MATLAB/C++ tutorials focused on computational fundamentals.

Awards and Scholarships

Outstanding PhD Student RA Scholarship, GIST	**2025**
Best Poster Award, Korean CDE Society	**2023–2025**
CDE DX Encouragement Award, Korean CDE Society	**2022**
Best Poster Award, Korean CDE Society	**2020–2021**
Outstanding PhD Student RA Scholarship, GIST	**2018–2021, 2023**
GIST Scholarship (Government support – Doctoral studies)	**2018–Present**
GIST Scholarship (Government support – Master's studies)	**2016–2018**

References

Prof. Kwang Hee Ko
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